

Zihao Fang

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EDUCATION

The Chinese University of Hong Kong, Shenzhen

September 2022 - June 2026

- **Bachelor of Engineering:** Computer Science and Engineering
- Cumulative GPA: **3.804/4.000**
- English Proficiency Test: **TOEFL 107**
- Courses Taken: Parallel Programming, Introduction to Embedded Systems, Compiler Construction, Cloud Computing, High-Performance Computing, Software Engineering, Design and Analysis of Algorithms, Network Programming, Machine Learning, Introduction to Cryptography, Blockchain and Decentralized Applications

Yale University

July 2023 - August 2023

- International Visiting Program
- Cumulative GPA: **4.000/4.000**
- Courses Taken: Discrete Mathematics, Introduction to Philosophy

PROJECTS

CUHK(SZ) Research Assistant Recruitment Website (Site-builder and Maintainer)

February 2023 - June 2025

- Building a dynamic website with Apache, PHP, and MySQL, along with all the maintenance jobs.
- Managing over 500 users, including over 100 Professors and 20 enterprise users.

RISC-V Elliptic Curve Cryptographic Acceleration Instruction Set Extension Design (Independent Researcher)

September 2023 - June 2025

- Design the hardware accelerator template with SystemVerilog, which is suitable for all elliptic curves in the Weierstrass form and prime field. Users can also customize their own cryptographic algorithms.
- Deconstruct the encryption and decryption processes, and find out primitive steps to simplify the instructions, therefore reducing the cost.

Optimization Attempt of Equity Structure Diagram Based on Artificial Intelligence (Project Director)

January 2024 - March 2024

- Use spiders to crawl publicly disclosed annual reports of companies in China, and use NLP algorithms to locate equity information pages in annual reports.
- Debug large model prompts, call ready-made large model APIs to analyze pages based on prompts, and generate JSON format equity information. After verification, the accuracy of large model information analysis and extraction is about 75%.
- Use Python to convert JSON information into SVG format equity structure diagrams to complete the project objectives.

GauU-Scene V2: Assessing the Reliability of Image-Based Metrics with Expansive Lidar Image Dataset Using 3DGS and NeRF (Research Assistant)

May 2024 - June 2024

Website: https://saliteta.github.io/CUHKSZ_SMBU/

- Optimize the algorithm used for dataset loading, use data chunking and lazy evaluation to optimize memory efficiency, and optimize the performance of the algorithm on machines with less memory. Use Python's multi-processing to parallelize the training and inference algorithms on the CPU, and use PyTorch to implement multi-GPU training and inference. After verification, the optimized parallel algorithm has a speedup ratio of up to 6.5.
- Build a GitHub Pages webpage for the GauU-Scene dataset results display to dynamically display the model of the dataset and the effect after processing.

INTERNSHIP

Alibaba Group Holding Limited — Hujing Digital Media & Entertainment Group (Operations Engineer)

July 2025 - September 2025

- Participate in Youku's daily stability operation guarantee and continuous high availability program landing.
- Participate in the architecture design and development of Youku middleware and the operation and maintenance platform.

CAMPUS LEADERSHIP & POSITION

SDS Student Club (student organization affiliated directly with the School of Data Science)

Head of Academic Department

May 2023 - May 2024

President

May 2024 - May 2025

- Organize numerous events and activities to improve the motivation of students studying fields related to computer science, data science, and statistics.
- Create the first interactive website of student organizations ever, and got the Creativity and Innovation Student Organization Award because of this.

Undergraduate Student Teaching Fellow

C/C++ Programming

September 2024 - May 2025

Parallel Programming

September 2025 - December 2025

Network Programming

January 2026 - May 2026

HONORS

Dean's List, 2023, 2024, and 2025 (awarded for outstanding academic performance)

CUHK(SZ) Programming Contest

- Second Prize, 2023
- Special Prize (Bronze), 2024

ASC Student Supercomputer Challenge

- Second Prize, 2025

The Academic Performance Scholarship

- Class C, 2024-2025

SKILLS

Programming Languages: C/C++, Java, Python, Verilog/SystemVerilog, SQL, Assembly

High-Performance Computing: pthread, OpenMP, MPI, CUDA, OpenACC, Triton, Slurm

Compiler: Lex, Yacc, Python-Lex-Yacc, LLVM, GCC

Others: Linux Cluster deployment and maintenance (including deployment with IPMI and ansible). Linux kernel modification and module insertion.

RESEARCH INTERESTS

Computer Architecture: RISC-V ISA Extension Design

Programming Language (Compiler): RISC-V LLVM Backend Design, LLVM Optimization Pass Design